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TITLE PAGE

- **Problem Statement ID** – 26166
- **Problem Statement Title-** Multi-modal, Sun Angle and Scale Invariant Image Correspondence using Chandrayaan-2 Optical Images (OHRC, TMC-2, IIRS)
- **Theme-** Space Technology
- **PS Category-** Software
- **Team ID-** SIH26-A0H-T109
- **Team Name-** HackRhythm

Hack Rhythm

Multi-modal, Sun Angle and Scale Invariant Image Correspondence using Chandrayaan-2 Optical Images (OHRC,TMC-2,IIRS)

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⚠The Problem, in Plain Language

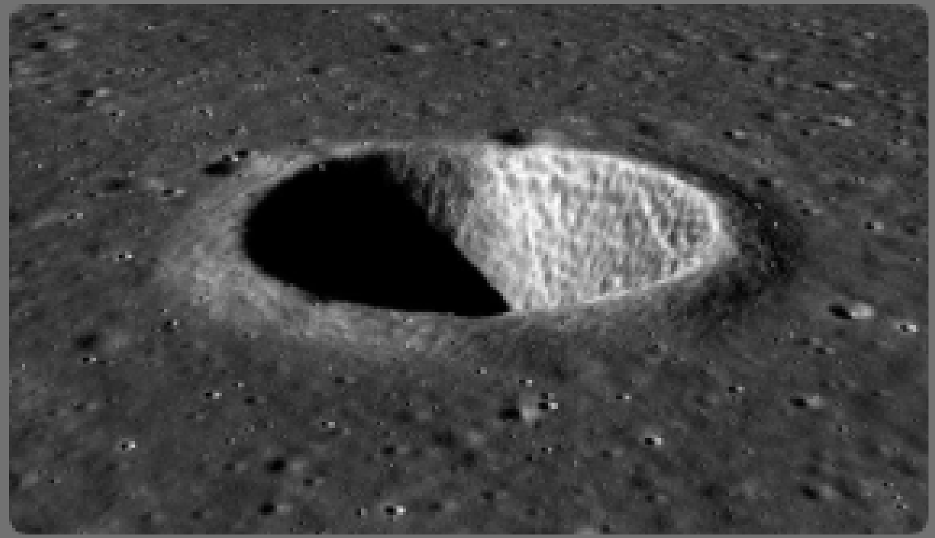
To detect water-ice and certify safe landing sites for Chandrayaan-4, ISRO must align three Chandrayaan-2 cameras into one master map. Existing software struggles because:

- 🔭 320× Zoom Gap — 1 IIRS pixel = 320 × 320 OHRC pixels, making shared landmarks difficult to find.
- 🌗 Shifting Polar Shadows — Low Sun angles change shadows between passes, confusing edge detection.
- 📡 Photo vs. X-Ray — OHRC uses optical imaging, while IIRS uses infrared sensing, making direct pixel matching unreliable.
- ⛰ 3D Terrain Drift — 3 km-high crater walls cause major distortion when treated as flat 2D terrain.

🎯 Planetary Geodetic Verification Framework

- **Validation Baseline:** Benchmarked against USGS Lunar Control Networks & LRO LOLA DEMs.
- **Tested Regimes:** Polar craters (Shackleton, Shoemaker) under extreme low-sun incidence (>75°).
- **Outlier Tolerance:** Stable with up to 70% feature occlusion in deep-shadow basins.
- **Ground Truth Method:** Manually verified tie-points on crater rims as the RMSE reference.

GeoPhase-Lunar Pipeline



- **SPICE Photometric Inversion:** Removes dynamic shadow bias using spacecraft orbital geometry.
- **Phase Congruency:** Illumination-invariant structural edges
- **RoMa/XoFTR:** Dense cross-sensor matching
- **Grid NMS, MAGSAC++, and TPS Refinement:** Sub-pixel geodetic refinement

Why We Stand Out

- **Staged Resolution Pyramids:** Bridges the **320x** scale gap via an intermediate TMC-2 step, keeping each match within proven scale ranges.
- **Sub-Pixel Geodetic Precision:** Targets **RMSE < 1.0px** for optical stages and **< 2.0px** for cross-modal pairs.
- **Graceful Degradation:** Automatic fallback chain (PCA → coarse align → manual seed) points if the primary matcher underperforms.

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TECHNICAL APPROACH

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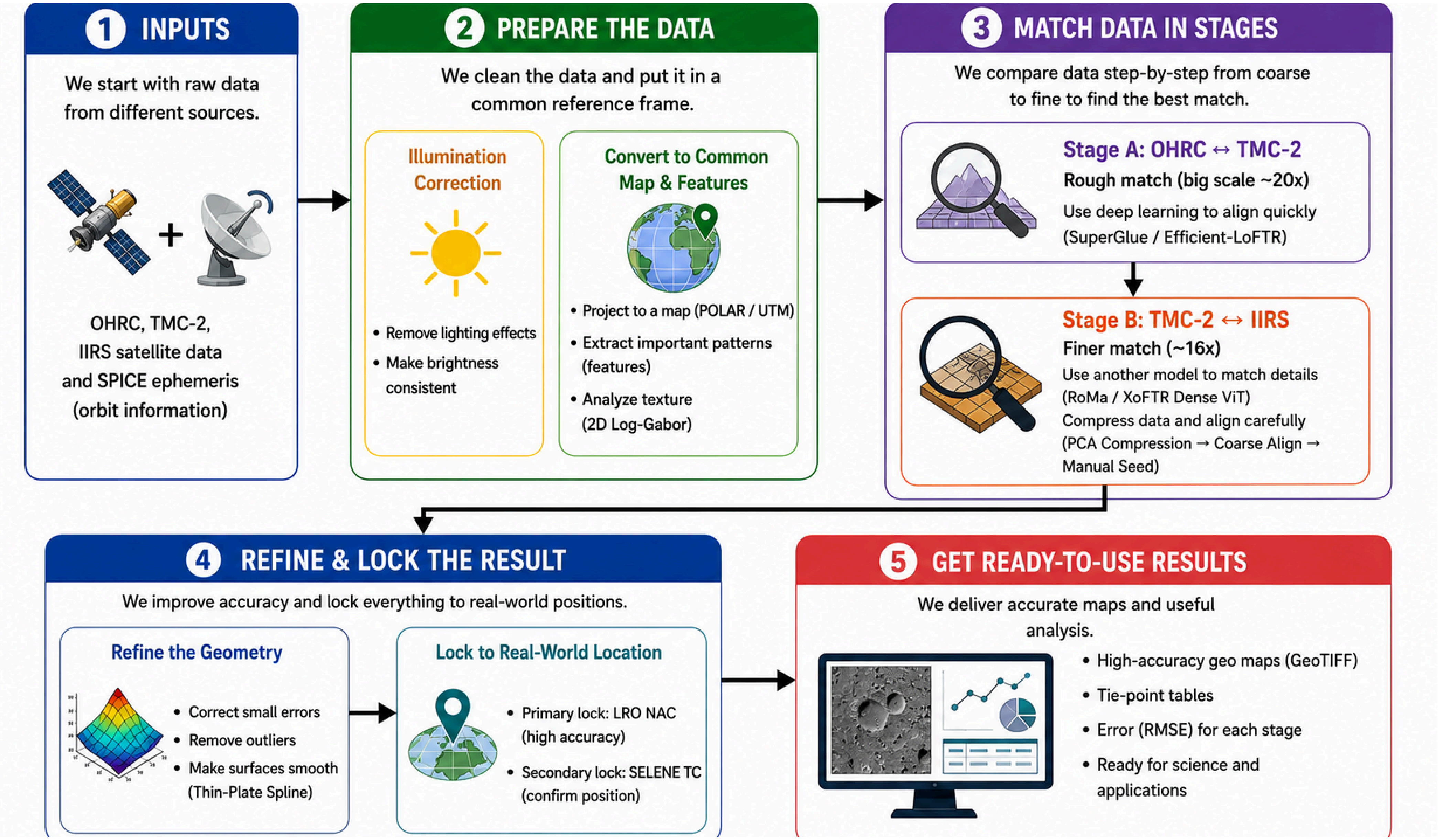

❖ Core & Geospatial: Python 3.10+, GDAL, and Rasterio for PDS4 ingestion and georeferencing; SpiceyPy for orbital SPICE kernels and solar geometry.

❖ Deep Learning & Matching: PyTorch and ONNX Runtime powering SuperGlue / Efficient-LoFTR (optical) and RoMa / XoFTR (cross-spectral matching).

❖ Computer Vision & Refinement: OpenCV (MAGSAC++) for outlier rejection, with SciPy for DEM-guided Thin-Plate Spline (TPS) warping.

❖ Frequency Analysis: CuPy and NumPy for accelerated 2D Log-Gabor Phase Congruency structural extraction.

❖ UI & Deployment: Streamlit and FastAPI for the interactive ground station demo, packaged via Docker for zero-cloud local execution.

Technical Stack:

INGESTION → GEOSPATIAL → MATCHING → REFINEMENT → UI → DEPLOY

python → GDAL → PyTorch → OpenCV → Streamlit → docker

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FEASIBILITY AND VIABILITY

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Feasibility:

The system is built to run entirely on standard laptops processing each image patch in under 1 second using less than 3.5 GB of GPU memory, with no cloud or server infrastructure required and integrates directly with official space data formats (PDS4/GeoTIFF) used by ISRO's PRADAN portal and NASA's LROC archive, grounding its results in optical physics rather than black-box AI.

Challenges / Risks	Strategy to Overcome
Extreme Shifting Shadows: at the lunar poles, low sun angles cast large moving shadows that trick normal computer vision into matching wrong shadow edges.	Physics-Based Shadow Removal (with Auto-Fallback): uses orbit metadata to mathematically remove shadows. If metadata is missing, it switches to contrast enhancement so it never fails.
Apples to Oranges: aligning a hyper-zoomed OHRC boulder photo with a low-res IIRS mineral map across a 320x zoom difference.	Stepping-Stone Strategy: align progressively instead of jumping the full 320x gap at once - High-Zoom (OHRC) -> Medium-Zoom (TMC-2) -> Hyperspectral (IIRS).
Rugged 3D Terrain: deep crater walls (>3 km) break normal flat 2D alignment and cause distortion/warping.	3D Elevation-Aware Alignment: uses lunar terrain height data to adjust for crater slopes, locking crater rims into place with sub-pixel precision.

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IMPACT AND BENEFITS

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Potential Impact on the Target Audience

- 🚀 **Reduces Manual Image Registration Effort (ISRO & Researchers):**Automates major stages of lunar image correspondence and alignment, reducing the time and effort required for manual tie-point selection and image preprocessing.
- 🌑 **Improves Lunar Mapping Under Difficult Conditions:** Enables more reliable matching across different Sun angles, resolutions, and sensor modalities, helping researchers analyse the same lunar region more consistently.
- 🛰️ **Supports Multi-Sensor Chandrayaan-2 Data Integration:** Helps bring OHRC, TMC-2, and IIRS data into a common correspondence framework, enabling combined geometric, visual, and spectral analysis.

Multi-Dimensional Benefits

- 💰 **Economic ROI & Cost Efficiency:** Saves millions in operational processing costs by creating an open, sovereign Indian mapping pipeline independent of expensive foreign commercial GIS software.
- 🇮🇳 **Strategic National Leadership:** Strengthens India's space software autonomy by providing a standardized, space-grade planetary mapping tool built for future lunar and planetary missions.
- 🛰️ **Direct Dual-Use on Earth (Disaster Management & Defense):** The core AI transfers directly to Earth observation — matching all-weather SAR with optical images for rapid flood mapping, cyclone damage assessment, and defense surveillance.
- ⚡ **Future Clean Energy & Space Prospecting:** Provides the spatial precision needed to map critical lunar resources — water-ice deposits, regolith titanium, and clean-energy reserves — for the next era of space exploration.

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RESEARCH AND REFERENCES

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